

CBK HOSTED KNET & T-PAY PAYMENT GATEWAY

Integration & Reference Manual

PAYMENT GATEWAY

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Payment Flow

A payment has following flow:

1) CREATION

Payment is created when a user (the customer) fills up and submits the payment information and it is sent to the CBK PG API. No processing has been done on the payment at this stage.

2) AUTHORIZATION & CAPTURE

An authorization is performed when user's payment details are successfully authenticated by the bank.

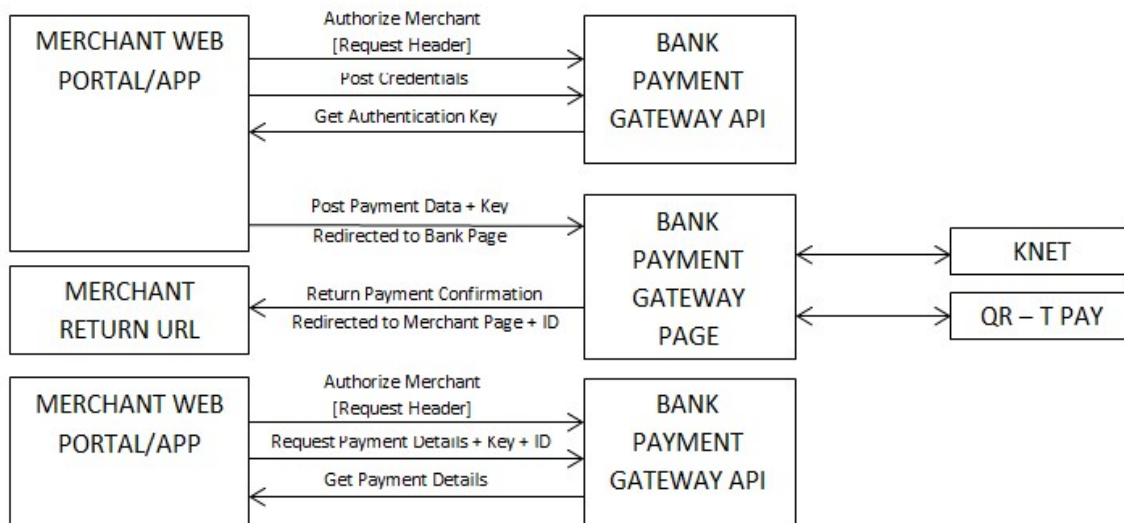
 **KNET:** The transaction amount is immediately deducted from the cardholder's account, and the corresponding credit to the merchant's account is processed on the following day (T+1).

K-Validate: KNET will verify the mobile number provided by the merchant against the mobile number registered with the issuing bank for the card being used. Merchants enrolled in the K-Validate service may include the customer's registered mobile number in the payment request parameters. For full details, refer to the "Request Parameters" section in the "Integration and Reference Manual"

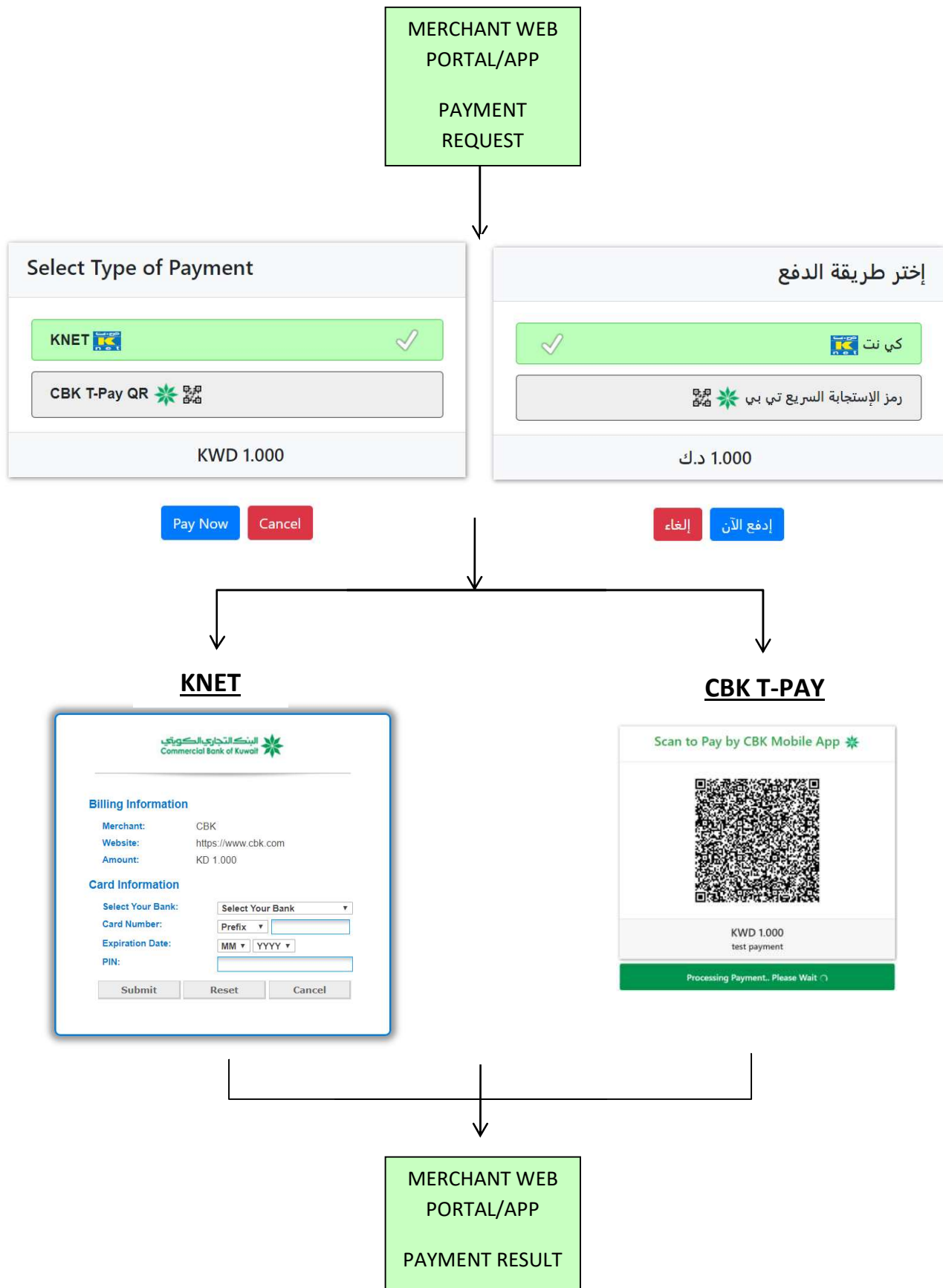
 **CBK T-Pay QR:** The amount is deducted from the customer's account (T-Pay Wallet), and will be transferred to the merchant's account (T-Pay Wallet) instantly.

RESULT

The user (customer) will be re-directed to the merchant's result page/return URL, after execution of payment. The detail of payment transaction is accessible by the PG API.



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Integration Guidelines & API Reference

Web services API Integration

Gateway Type: Hosted Gateway

Communication Protocol: HTTPS

Architectural Style: REST (Representational State Transfer)

Format: JSON (JavaScript Object Notation)

Protocol: REST-JSON

Prerequisite

SSL/TLS: Required

Security Protocol: TLS 1.2

Result URL: Merchant Payment result URL

Server Static IP: API accessing server IP. Max 2 IPs allowed. *

** Optional. Merchants with Static IP will have IP filtered Server connection to API access*

Authentication Key

To authenticate and to gain access to resources, you use personal access token (Authenticate Key) generated by the gateway API. Tokens are similar to passwords. A token expires and can be revoked. It is to be used in all API access to authenticate merchant access.

Authorization:

Authorization header will authorize the merchant access to API and Authenticate Key verifies the access to the process.

All server side requests as generating authentication key, creating transaction request and getting payment details must be accessed with Authorization header & Authenticate key along.

Protocol: REST-JSON

Authorization Header: HTTP header

Authorization Header Type: Basic (with base64-encoded)

Basic Auth: Scheme that uses an encoded username and password for credentials

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Format: "Basic {ClientId}:{ClientSecret}"

Example: "Basic 458224558:fqzYHEV34lk3409dfsfs45bGsdfk345lkmE2R0bw"

AUTHORIZATION ENTITY REFERENCE

Attribute	Description	Data Length (Max)
ClientId	Merchant API ID	50
ClientSecret	Merchant API Password	100

Generating Authenticate Key

REQUEST:

The Authenticate Key (AccessToken) is generated or received by posting data to Authenticate API URL. Authenticate key will expire after 6* hours from time of generation. Save the key locally and re-use till expiry. Regenerate new Authenticate key to access any function again on failure.

Test URL: {TestPG}/ePay/api/cbk/online/pg/merchant/Authenticate

Production URL: {ProductionPG}/ePay/api/cbk/online/pg/merchant/Authenticate

Protocol: REST-JSON

Request Method: POST

Content-Type: application/json

Authorization: required

AUTENTICATION ENTITY REFERENCE

JSON Post Data

Attribute	Description	Data Length (Max)
ClientId	Merchant API ID	50
ClientSecret	Merchant API Password	100
ENCRP_KEY	Merchant Encrypted account key	500

Your {ClientId}, {ClientSecret} & {ENCRP_KEY} are never to be used for client side requests and never to share it with anyone.

Please contact bank support to reset/change the {ClientSecret} or {ENCRP_KEY}

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Example:

```
{
  "ClientId":"458224558",
  "ClientSecret":"fqzYHEV34Ik3409dfsfs45bGsdfIk345IkmE2R0bw",
  "ENCRP_KEY":"UvpnachOpyUc7iKw9g_c8T9OPyckwGqNcVzq5SlgDS-P3zcvvGavT16"
}
```

RESPONSE:

Response Format: JSON

HTTP STATUS CODE REFERENCE

HTTP Status Code	Description
200 OK	Worked with result
400 Bad Request	Bad request to API
401 Not Authorized	Authentication Error. Invalid {ClientId} or {ClientSecret} or Invalid IP
404 Not Found	Error / Not Found
500, 502, 504	Gateway Error. Internal PG Error

RESPONSE ENTITY REFERENCE

Successful

Attribute	Description	Data Length (Max)
Status	Status of Authentication	2
AccessToken	Authentication Key	100

Example:

```
{
  "Status":"1",
  "AccessToken":"cecHAgtwSUNuf2PK4YUVNQubi6mReFEoNL7BmotBh7E1"
}
```

Failed

Attribute	Description	Data Length (Max)
Status	Status of Authentication	2
Message	Failed Message	100

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Example:

```
{
  "Status":"0",
  "Message":"Invalid"
}
```

STATUS CODE REFERENCE

Status Code	Message	Description
0	Invalid	Access Invalid. API Authentication failed
1	Success	{AccessToken} Authentication key returned

Gateway Checkout

The hosted checkout allows collecting payment details and sending to the hosted URL of Payment Gateway for payment processing. The payment status will be checked and updated to merchant upon execution to result URL.

Protocol: NVP (Name-Value Pair)

Payment Request

REQUEST:

Operations are sent to the gateway as HTTP POST operations with all arguments as name-value pairs (NVP) in the HTTP body. The encoding is “application/x-www-form-urlencoded” as defined in HTML4.01.

Test URL: {TestPG}/ePay/pg/epay? v={AccessToken}

Production URL: {ProductionPG}/ePay/pg/epay? v={AccessToken}

URL Parameter: _v={AccessToken}

Request Method: POST

Content-Type - application/x-www-form-urlencoded

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REQUEST PARAMETERS:

Field Name (NVP Parameters)	Description	Required?	Data Type
tij_MerchantEncryptCode	Encrypted merchant permanent account key	Yes	Max 500
tij_MerchAuthKeyApi	Authentication Key {AccessToken} – Encrypted	Yes	Max 100
tij_MerchantPaymentLang	English/Arabic	Yes	en/ar
tij_MerchantPaymentAmount	Payment Amount	Yes	Numeric Max 10 digit (including 3 decimals)
tij_MerchantPaymentTrack	Unique Merchant Transaction Track/Order ID. Will be returned with payment details to identify the transaction	Yes	Alphanumeric – Max 30 (Unique)
- Alpha and Numeric. Include uppercase and lowercase characters with numeric. - Sample: ABCabc123			
tij_MerchantPaymentRef	Short Payment Description to be displayed with payment T-Pay QR page	Optional	Alphanumeric – Max 30
- English & Arabic Alpha, Numeric. Include uppercase and lowercase characters with numeric and special characters (-.&) and space - Sample: ABC&abc-1.23			
tij_MerchantPaymentCurrency	KWD (Kuwait Dinar) - Default	Optional	KWD
tij_MerchantUdf1	Merchant defined value to be returned	Optional	Alphanumeric – Max 20
- Alpha and Numeric. Case sensitive and include uppercase and lowercase characters with numeric - Sample: ABCabc123			
tij_MerchantUdf2	Merchant defined value to be returned. This value will be taken as reference into the Bank transaction statement.	Optional	Alphanumeric – Max 10
- Alpha, Numeric which Include uppercase and lowercase characters with special character hyphen (-) and space - Sample: ABC-abc 123			
tij_MerchantUdf3	Merchant defined value to be returned	Optional	Alphanumeric – Max 100

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- Alpha, Numeric which Include uppercase and lowercase characters with special character hyphen (-) and space - Sample: ABC-abc 123			
tij_MerchantUdf4	Merchant defined value to be returned – OR – KNET K-Validate data *	Optional	Alphanumeric – Max 100 – OR – Defined Format*
- Alpha, Numeric which Include uppercase and lowercase characters with special character hyphen (-) and space - Sample: ABC-abc 123 - * K-Validate format - “CV” + “space” + “Customer Mobile number” - * Sample: “CV 99668899”			
tij_MerchantUdf5	Merchant defined value to be returned	Optional	Alphanumeric – Max 100
- Alpha, Numeric which Include uppercase and lowercase characters with special character hyphen (-) and space - Sample: ABC-abc 123			
tij_MerchPayType *	Payment Mode – Direct Access (No Payment selection page) (See Payment Code below)	Optional	Numeric - 2
tij_MerchReturnUrl	Merchant Return URL without URL Parameters	Yes *	Valid URL – Max 200
- Valid Web Secured URL (https://) - * Optional for Merchants with Static URL registered with gateway			

Payment Mode Reference *:

Code	Description	Currency Support
1	KNET	KWD
2	CBK T-Pay QR	KWD

Currency Code Reference **:

Code	Description	Decimal (Example)
KWD	Kuwait Dinar	3 (100.000)

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Example:

```
<form id="form1" method="post" action= {Checkout URL}"
  enctype="application/x-www-form-urlencoded">
  <input type="hidden" name="tij_MerchantEncryptCode" value="{ENCRP_KEY}" />
  <input type="hidden" name="tij_MerchAuthKeyApi" value="{AccessToken}" />
  <input type="hidden" name="tij_MerchantPaymentLang" value="en" />
  <input type="hidden" name="tij_MerchantPaymentAmount" value="1" />
  <input type="hidden" name="tij_MerchantPaymentTrack" value="123" />
  <input type="hidden" name="tij_MerchantPaymentRef" value="Description" />
  <input type="hidden" name="tij_MerchantPaymentCurrency" value="KWD" />
  <input type="hidden" name="tij_MerchantUdf1" value="1" />
  <input type="hidden" name="tij_MerchantUdf2" value="" />
  <input type="hidden" name="tij_MerchantUdf3" value="" />
  <input type="hidden" name="tij_MerchantUdf4" value="CV 99668899" />
  <input type="hidden" name="tij_MerchantUdf5" value="" />
  <input type="hidden" name="tij_MerchPayType" value="" />
  <input type="hidden" name="tij_MerchReturnUrl"
value="https://www.xxxxx.com/response.php" />

  <input type="submit" value="Submit" />
</form>
```

Note:

- Merchant Track/Order ID must be always unique for each transaction attempts from merchant.
- Get updated {AccessToken} from gateway server before checkout to payment gateway.

Payment Result

After completion of a transaction cycle, the gateway will return to merchant Re-direct URL (provided during setup or sent in URL parameter) and the transaction result data can be accessed as given below. The Merchant MUST save the returned values by PG in their database.

Result Page (Merchant Re-direct URL) will be returned with an encrypted variable value.

Response Parameter	Description
encrp	Unique Encrypted URL GET Parameter value for each transaction

Example: <https://{returnurl}?{encrp}=value>

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Transaction Result:

The transaction result can be retrieved by two different APIs with authentication

REQUEST:

1. RETURN URL WITH RESPONSE PARAMETER

The transaction result can be returned by accessing the API with the encrypted returned parameter {encrp} and Authenticate token to the gateway

Protocol: REST-JSON

Request Method: GET

Authorization: required

Test URL: {TestPG}/ePay/api/cbk/online/pg/GetTransactions/{encrp}/{AccessToken}

Production URL:

{ProductionPG}/ePay/api/cbk/online/pg/GetTransactions/{encrp}/{AccessToken}

REQUEST ENTITY REFERENCE

Parameter Variable	Description
encrp	Encrypted URL GET Parameter
AccessToken	Authentication Key

2. INQUIRE/VERIFY WITH TRACK ID

The transaction result can be returned by posting the unique merchant track/order id with Authenticate token and account key to the gateway.

Protocol: REST-JSON

Request Method: POST

Content-Type: application/json

Authorization: required

Test URL: {TestPG}/ePay/api/cbk/online/pg/Verify

Production URL: {ProductionPG}/ePay/api/cbk/online/pg/Verify

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REQUEST ENTITY REFERENCE

Attribute	Description
encrypmerch	Merchant Encrypted account key
authkey	Authentication Key
payid	Unique merchant track/order id

RESPONSE:

Response Format: JSON

HTTP STATUS CODE REFERENCE

HTTP Status Code	Description
200 OK	Worked with result
400 Bad Request	Bad request to API
401 Not Authorized	Authentication Error. Invalid {ClientId} or {ClientSecret} or Invalid IP
404 Not Found	Error / Not Found
500, 502, 504	Gateway Error. Internal PG Error

PAYMENT DETAILS ENTITY REFERENCE

Successful Result

Attribute	Description	Data Length/Size (Max)
Status	Payment Status *	2
Message	Payment Message *	50
Amount	Amount of transaction	11
AuthCode	Auth Code by Bank [KNET]	20
PaymentId	Payment ID [KNET]	20
PayType	Type Of Payment Channel	20
ReceiptNo	Receipt No	20
ReferenceId	Reference ID [KNET]	20
TrackId	Payment Gateway Track ID	30
TransactionId	Transaction ID [KNET][QR]	20
PayId	Merchant Track ID	30
MerchUdf1	Merchant Extra Variable	20
MerchUdf2	Merchant Extra Variable	10
MerchUdf3	Merchant Extra Variable	100

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MerchUdf4	Merchant Extra Variable	100
MerchUdf5	Merchant Extra Variable	100

* Refer Payment Status Code reference

Example:

```
{
  "Status":"1",
  "Message":"Success",
  "Amount":"1",
  "AuthCode":"",
  "PaymentId":"",
  "PayType":"KNET",
  "ReceiptNo":"12312",
  "ReferenceId":"",
  "TrackId":"123123123",
  "TransactionId":"",
  "PayId":"123",
  "MerchUdf1":"1",
  "MerchUdf2":"",
  "MerchUdf3":"",
  "MerchUdf4":"",
  "MerchUdf5":""
}
```

Failed Result

Attribute	Description
Status	Status of Authentication
Message	Failed Message

Example:

```
{
  "Status":"0",
  "Message":"Invalid"
}
```

Note: If Status=0, re-generate new API key and try again.

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Error

On any error during the transaction, the result page (Merchant Re-direct URL) will be returned with the mentioned parameter variable values

Parameter Variable	Description
ErrorCode	Error Code value. Please refer to the Error Codes table for further details
PayTrackID	Unique Merchant Transaction Track/Order ID

Example: `https ://{returnurl}?{ ErrorCode}=value&{PayTrackID}=value`

TRANSACTION TYPE REFERENCE *

PayType	Description
KNET	KNET Transaction
QR	CBK T-Pay QR

PAYMENT RESULT STATUS CODE REFERENCE *

Status Code	Message	Description
-1	Invalid	Error/Invalid {encrp}/{payid}
0	Invalid	Invalid Access. Re-generate new API key
1	Success	Payment Successfully Authorized & Captured
2	Failed	Payment Failed
3	Expired/Cancelled	Payment Cancelled or Expired

*The Merchant **MUST** save the returned values by PG in their database.*

The QR code page should not be closed by the user if the payment is being executed on the QR to receive correct transaction results.

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ERROR CODES

Code	Description
TIJ0001	Invalid Merchant Language
TIJ0002	Invalid Merchant Amount
TIJ0003	Invalid Merchant Amount KWD
TIJ0004	Invalid Merchant Track ID
TIJ0005	Invalid Merchant UDF1
TIJ0015	Invalid Merchant UDF2
TIJ0022	Invalid Merchant UDF3
TIJ0023	Invalid Merchant UDF4
TIJ0024	Invalid Merchant UDF5
TIJ0027	Invalid Merchant Return URL
TIJ0006	Invalid Merchant Currency
TIJ0007	Invalid Merchant Payment reference
TIJ0008	Invalid Merchant Pay Type
TIJ0009	Invalid Merchant API Authenticate Key
TIJ0016	Error in QR
TIJ0020	Error in KNET
TIJ0036	Error in TCASH
TIJ0010	Invalid Access
TIJ0017	Invalid Page Access
TIJ0011	Invalid Merchant Key
TIJ0012	Duplicate Merchant Track ID
TIJ0013	Invalid Merchant Key (not match)
TIJ0014	Invalid Merchant Key (not available)
TIJ0019	Invalid KNET/QR Data
TIJ0021	Error Processing Data
TIJ0031	Transaction session expired